

Review retrospective:
*the potential-agnostic affine κ_3 identity — the abstraction its
harmonic and anharmonic specialisations rest on*

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Abstract

A soundness review of `Laplace/AffineKappa3.lean`: the potential-agnostic factored third-cumulant identity $\kappa_3(L, a_2x + b_2, t, a_1x + b_1, a_3x + b_3) = a_1a_2a_3 \kappa_3(L, x, t, x, x)$ — the general result the harmonic-vanishing and anharmonic-asymptotic κ_3 files specialise. Result: a clean fidelity pass with two small edits (a dead equation name and a misleading-docstring clarification).

1 Setting

This is the abstraction beneath `HarmonicAffineKappa3` and `AnharmonicKappa3Affine`: rather than re-derive the affine factoring per potential, it proves once, for an arbitrary potential L (given $Z \neq 0$ and the degree- ≤ 3 moment integrabilities), that shifting and scaling the three observables of κ_3 multiplies the all-identity cumulant by $a_1a_2a_3$ and drops the additive constants.

2 What was reviewed

The Threepoint $\rightarrow$ Laplace bridge `threepoint_gibbsExp_volume_zero.eq`; the five-term unfold `kappa3_volume_unfold`; the affine and cubic expectation reductions `gibbsExpectation_affine` and `gibbsExp_cubic.eq`; and the headline `kappa3_volume_shifted_affine.eq_smul` — checked for correct cumulant signs, polynomial coefficients, and the genuine cancellation of the b -constants.

3 Fidelity / soundness findings

Sound; no defect. The reviewer confirmed the five-term unfold’s signs (+ − − − +), every coefficient in the seven product-term expansions, and the headline $a_1a_2a_3$, and then verified the *cancellation* directly: after substituting the moments the left side collapses to $a_1a_2a_3 (m_3 - 3m_2m_1 + 2m_1^3) = a_1a_2a_3 \kappa_3(x, x, x)$, with every b -dependent contribution — the one- b m_2 terms, the two- b m_1 terms, the one- b m_1^2 terms, and the pure constant $b_1b_2b_3$ (whose coefficients sum $+1-1-1-1+2 = 0$) — summing to zero. “No mathematical/algebraic fidelity defects.”

4 Simplifications and documentation fixes applied

A dead equation name and a misleading docstring. The first-moment `set ...with hm1` carried an equation name `hm1` that — unlike `hm2`/`hm3`, which are rewritten with by name — is never

used; dropped (the abbreviation it names stays). And `kappa3_volume_unfold`’s docstring claimed the unfolded expression is “independent of the perturbation direction A ,” which reads false: A is plainly all over the right-hand side as an *observable* (φAB , AB , A). The accurate point — which the fix now states — is that A drops only from the Gibbs *weight* at $h = 0$ (the perturbation $h \cdot A$ leaves the integrand), which is what lets the unfold go through the *unperturbed* `gibbsExpectation`; A survives as an observable. Build green, no warnings, signatures byte-identical; the reviewer affirmed both.

5 Disagreements and open questions

None. The file opens only `MeasureTheory` (no dead open `Real` to shed — already minimal). The `import ...Bochner.Basic` is redundant (`Laplace.Gibbs` supplies it transitively) but its symbols `Integrable/Measure/volume` are used directly, so it was kept — redundant-but-used is not dead code (the same call made for the redundant `Anharmonic` import in the `IntegralRemainder` review).

6 Lessons

- **An “independent of X ” docstring is worth a second look when X is visibly in the statement.** The claim was true of the Gibbs *weight* but false of the *expression*; the fix is to name which object is actually X -independent. A reader who took the docstring literally would have been misled about what the unfold says.
- **A potential-agnostic lemma is the right place to verify the cancellation once.** Because this file proves the affine factoring for arbitrary L , confirming the b -drop here certifies it for every specialisation (harmonic, anharmonic, ...) at once — the review analogue of the abstraction the file itself provides.

7 Follow-ups

- With this, the only never-reviewed laplace files left are the large `Multi/` analytic files: `RescaledIntegrals` (1547L) and the `Covariance/CovarianceSharp/ CovarianceExplicit` trio (up to 19.6k lines), which want sectioned reviews.